Utanos: A general-purpose shallow whole genome sequencing analysis toolkit identifies interpretable copy number signatures in endometrial cancer

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Background & rationale

Background:

- Genomic copy-number (CN) variation is a feature and driver of many cancers, notably ovarian cancer and endometrial cancer.
- Whole-genome sequencing (WGS) is the gold standard technique for assessing genomic variation in cancer.

Rationale:

- The cost of WGS is prohibitive for many applications, but shallow whole-genome sequencing (sWGS) can profile genomic CN variation at a fraction of the cost.
- Tool which integrate the generation of CN profiles with downstream analyses to identify biological significance are lacking.

Aims:

- We sought to develop general-purpose R-package for the analysis of CN-profiles: UTANOS (UTilities for the ANalysis Of Shallow WGS).
- We applied Utanos to 196 p53-abnormal endometrial cancer samples, identifying 5 copy-number signatures associated with distinct mechanisms of genomic alteration, stratifying samples into 5 distinct clinically relevant groups (Jamieson et al., 2024).

The Utanos package

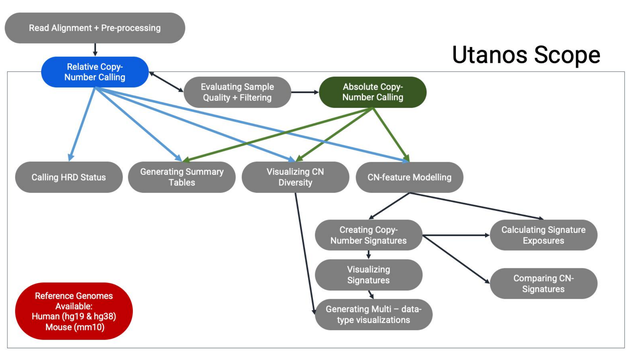


Figure 1. The scope of the Utanos package. Prior to using Utanos, raw reads should first be aligned to the desired reference genome, pre-processed (ex. duplicate marking), and relative copy-numbers called.

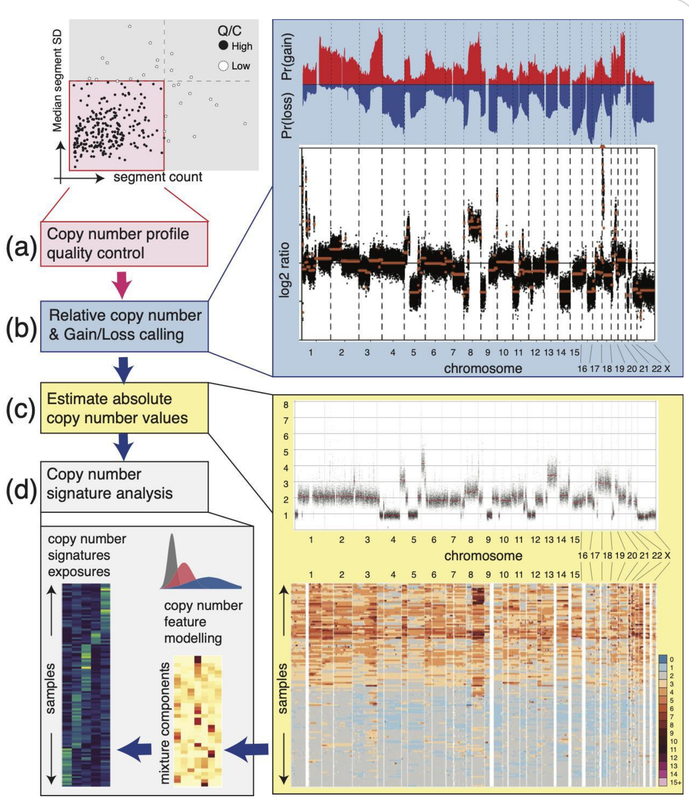


Figure 2. The Utanos workflow, from quality control to copy-number signatures.

Results

Key oncogenes and tumor repressors are disrupted at the copy-number level in p53-abnormal endometrial cancer

Endometrial cancer:

- In 2013, the Cancer Genome Atlas (TCGA) identified four molecularly distinct subtypes of endometrial cancer: POLE ultramutated, microsatellite instability ultramutated, CN low, and CN high.
- The CN high subtype is associated with <50% 5-year survival, many somatic copy-number alterations, and few mutations, except for many TP53 mutations.
- Abnormal P53 staining on IHC characterizes the CN-high group.

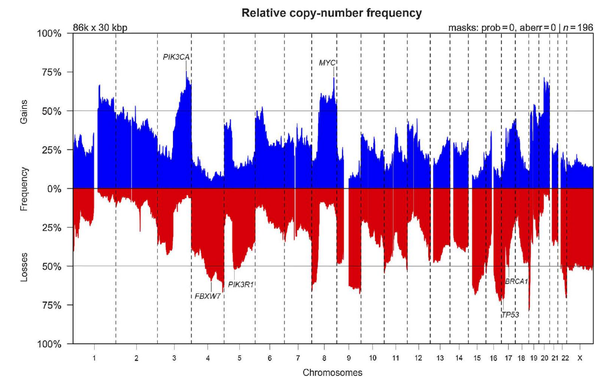


Figure 3. Frequency of relative copy-number (rCN) gains (log2(rCN) > 0.1) and losses (log2(rCN) < -0.1) in 196 p53-abnormal endometrial cancer samples. Previously published in Jamieson et al., 2024.

Estimation of tumor content (cellularity) and absolute copy-number

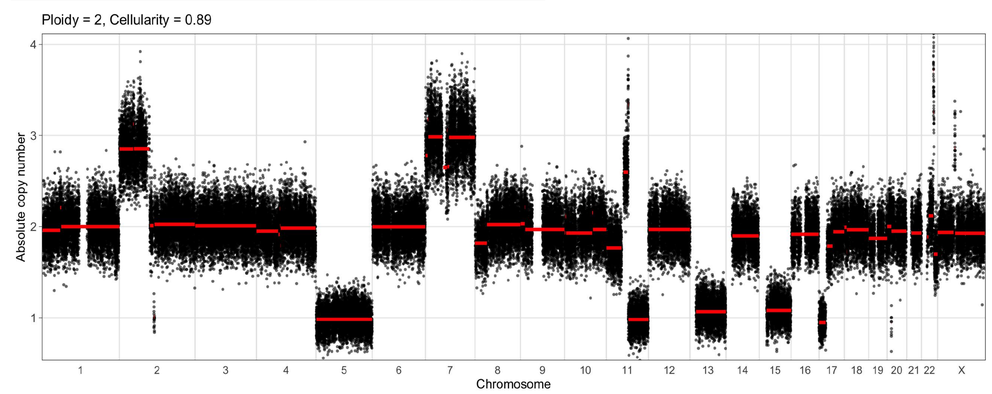
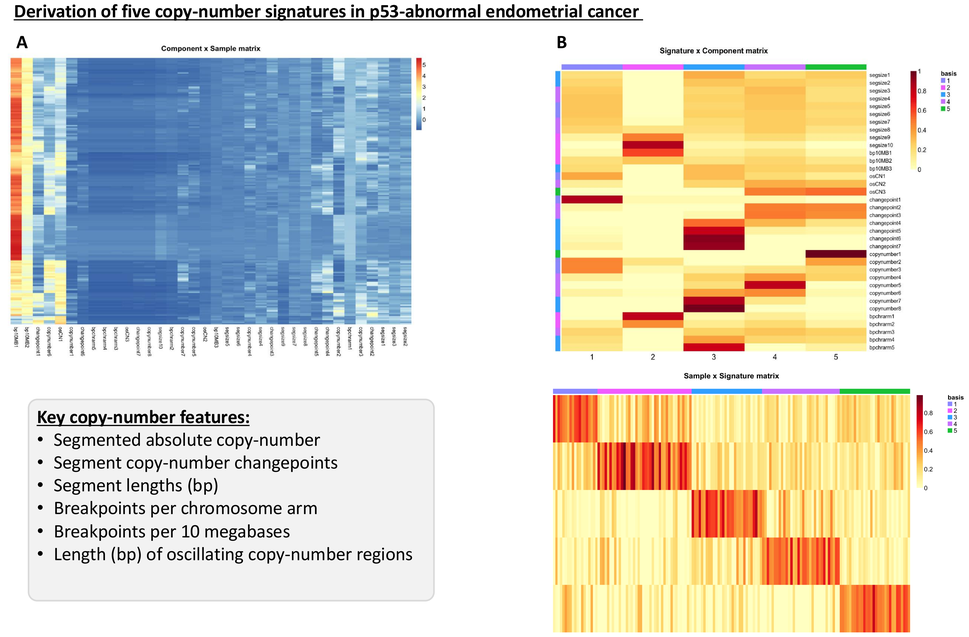


Figure 4. A representative example of the estimation of absolute CN (aCN) from relative CN. Utanos estimates tumor ploidy and cellularity using various methods implemented by the rascal R-package. In this case, a grid-search of possible values of ploidy and cellularity is conducted. The combination of ploidy and cellularity with the lowest median absolute deviation in segment aCN from the nearest integer is selected as the most likely solution and plotted.

Derivation of five copy-number signatures in p53-abnormal endometrial cancer



Key copy-number features:

- Segmented absolute copy-number
- Segment copy-number changepoints
- Segment lengths (bp)
- Breakpoints per chromosome arm
- Breakpoints per 10 megabases
- Length (bp) of oscillating copy-number regions

Figure 5. A) Features are extracted from the aCN profiles of each sample. Using the flexmix R-package (Grün and Leisch 2008), Utanos models each feature as a mixture of either Gaussian or Poisson distributions, yielding components. For each sample, the posterior probability of each event belonging a component is computed and summed; the resulting vectors are combined, giving the component x sample matrix. **B)** Via the NMF R-package (Gaujoux and Seoighe 2010), Utanos non-negatively factorizes the component x sample matrix into the signature x component matrix and sample x signature matrix; functions and visualizations are provided to determine the optimal number of signatures.

Prediction of homologous recombination deficiency (HRD) from rCN

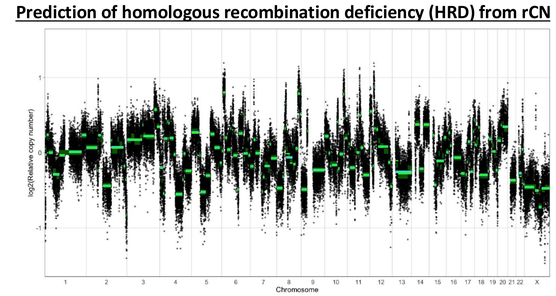


Figure 6. Example of the prediction of HRD from rCN for a sample predicted to have HRD. Utanos implements the algorithm known as shallowHRD (Eeckhoutte et al., 2020). HRD is predicted when the number of large genomic alterations (segments > 10 Mb) exceeds 19. To do so, the initial segments are merged in two passes based on a threshold estimated from the data using kernel density estimation. The initial segments are given in dark green; segments merged in the first pass are turquoise; the final segmentation is bright green.

Copy-number signatures are associated with distinct genomic events

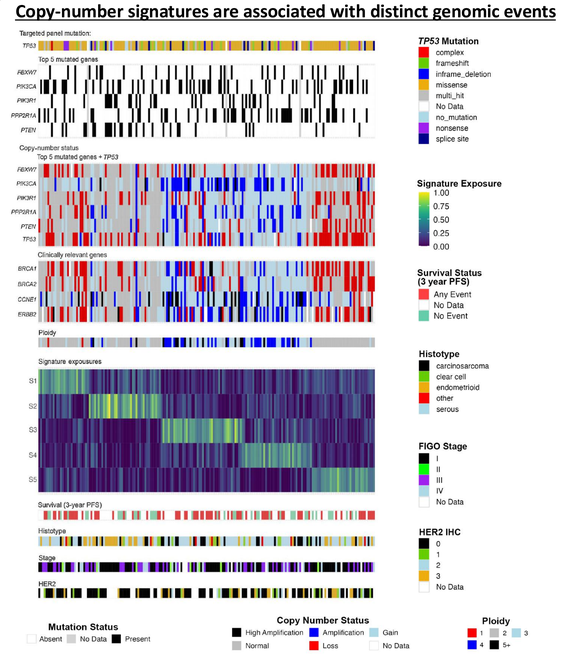


Figure 7. Plot summarizing the relationship between signature exposures per sample in the p53-abnormal endometrial cohort (n = 187), targeted panel mutations, copy-number status, and various clinical correlates. Previously published in Jamieson et al., 2024. Targeted panel sequencing was conducted separately from sWGS.

Conclusions

Utanos provides functions to comprehensively characterize sWGS data:

- Quality assessment
- Visualization of per-sample and entire-cohort copy-number profiles
- Estimation of absolute copy-number
- Extraction and visualization of copy-number features; mixture-modeling
- Derivation and visualization of copy-number signatures
- Prediction of homologous recombination deficiency from relative copy-number
- See: https://github.com/Huntsmanlab/utanos

Through our analysis of 196 p53-abnormal endometrial cancers, we show that sWGS is a clinically relevant and cost-effective alternative to WGS for evaluating CN-variation and its significance.

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